

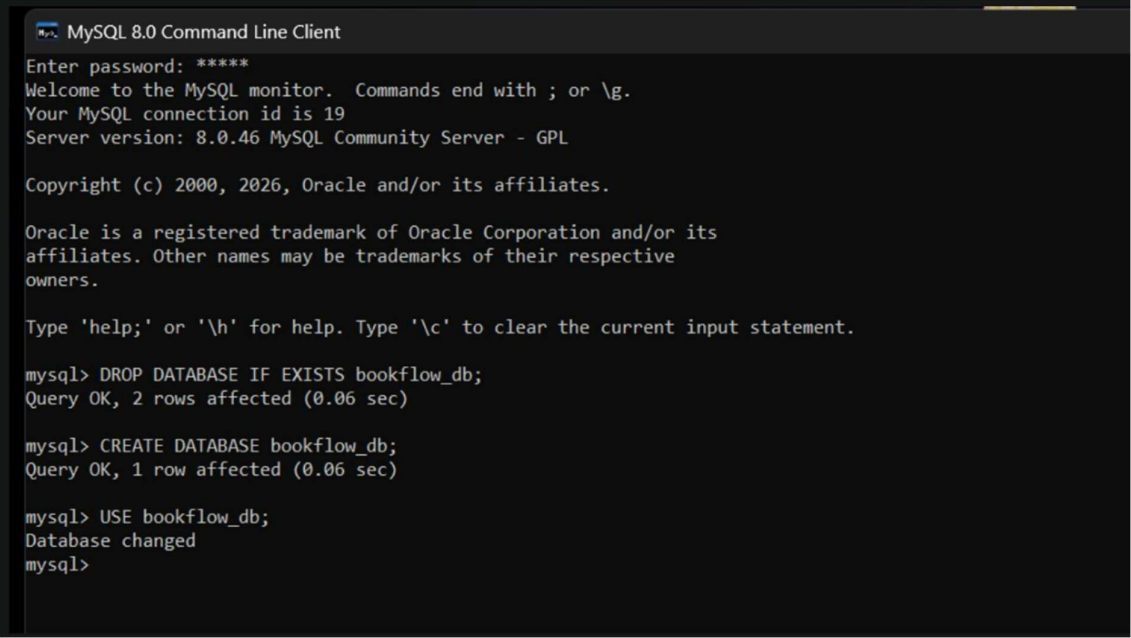
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

Step 1 :Database Setup Query 1: Create the database

CREATE DATABASE IF NOT EXISTS bookflow_db;

A screenshot of the MySQL 8.0 Command Line Client interface. The window title is "MySQL 8.0 Command Line Client". The prompt "Enter password: *****" is shown. Below it, a welcome message states: "Welcome to the MySQL monitor. Commands end with ; or \g. Your MySQL connection id is 19. Server version: 8.0.46 MySQL Community Server - GPL. Copyright (c) 2000, 2026, Oracle and/or its affiliates. Oracle is a registered trademark of Oracle Corporation and/or its affiliates. Other names may be trademarks of their respective owners. Type 'help;' or '\h' for help. Type '\c' to clear the current input statement." The user enters the command "mysql> DROP DATABASE IF EXISTS bookflow_db;" and receives the response "Query OK, 2 rows affected (0.06 sec)". Then, the user enters "mysql> CREATE DATABASE bookflow_db;" and receives "Query OK, 1 row affected (0.06 sec)". Finally, the user enters "mysql> USE bookflow_db;" and receives "Database changed", followed by the "mysql>" prompt.

```
MySQL 8.0 Command Line Client
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 8.0.46 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql>
```

Query 2: Select the database

USE bookflow_db;

```
MySQL 8.0 Command Line Client
Your MySQL connection id is 19
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->   book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   title VARCHAR(255) NOT NULL,
  ->   isbn VARCHAR(13) NOT NULL UNIQUE,
  ->   published_year INT,
  ->   CONSTRAINT chk_published_year
  ->   CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

3. Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table

```
CREATE TABLE books (
  book_id    INT AUTO_INCREMENT PRIMARY KEY,
  title      VARCHAR(255) NOT NULL,
  isbn       VARCHAR(13) NOT NULL UNIQUE,
  published_year INT,
  CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
```

```
MySQL 8.0 Command Line Client
Your MySQL connection id is 19
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->     book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->     title VARCHAR(255) NOT NULL,
  ->     isbn VARCHAR(13) NOT NULL UNIQUE,
  ->     published_year INT,
  ->     CONSTRAINT chk_published_year
  ->     CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

Query 4: Create the members table

```
CREATE TABLE members (
  member_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE
);
```

```
MySQL 8.0 Command Line Client
-> published_year INT,
-> CONSTRAINT chk_published_year
-> CHECK (published_year < 2027)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE members (
-> member_id INT AUTO_INCREMENT PRIMARY KEY,
-> full_name VARCHAR(100) NOT NULL,
-> email VARCHAR(150) NOT NULL UNIQUE
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | auto_increment |
| title | varchar(255) | NO | | NULL | |
| isbn | varchar(13) | NO | UNI | NULL | |
| published_year | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql>
```

Query 5 (verification): Inspect the table structures

```
MySQL 8.0 Command Line Client
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | auto_increment |
| title | varchar(255) | NO | | NULL | |
| isbn | varchar(13) | NO | UNI | NULL | |
| published_year | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> DESCRIBE members;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id | int | NO | PRI | NULL | auto_increment |
| full_name | varchar(100) | NO | | NULL | |
| email | varchar(150) | NO | UNI | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

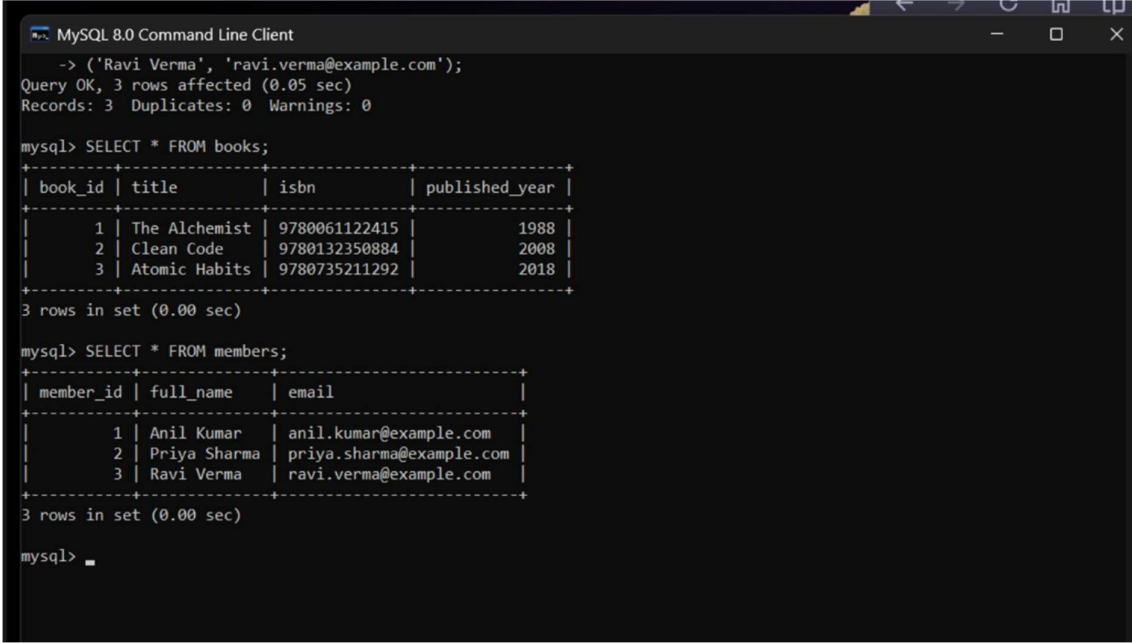
mysql>
```

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Anil Kumar', 'anil.kumar@example.com'),
('Priya Sharma', 'priya.sharma@example.com'),
('Ravi Verma', 'ravi.verma@example.com');
```



```
MySQL 8.0 Command Line Client
-> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM books;
+-----+-----+-----+-----+
| book_id | title       | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988           |
| 2       | Clean Code   | 9780132350884 | 2008           |
| 3       | Atomic Habits | 9780735211292 | 2018           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name | email                     |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com   |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma  | ravi.verma@example.com   |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Explanation: The ISBN 9780061122415 already belongs to 'The Alchemist'. The UNIQUE constraint on isbn rejects the row, so two physical records can never claim the same ISBN.

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Explanation: The NOT NULL constraint fires because a book must have a name. The database refuses to store a nameless book, exactly as the requirement demands.

Test 3: Future publication year (CHECK violation)

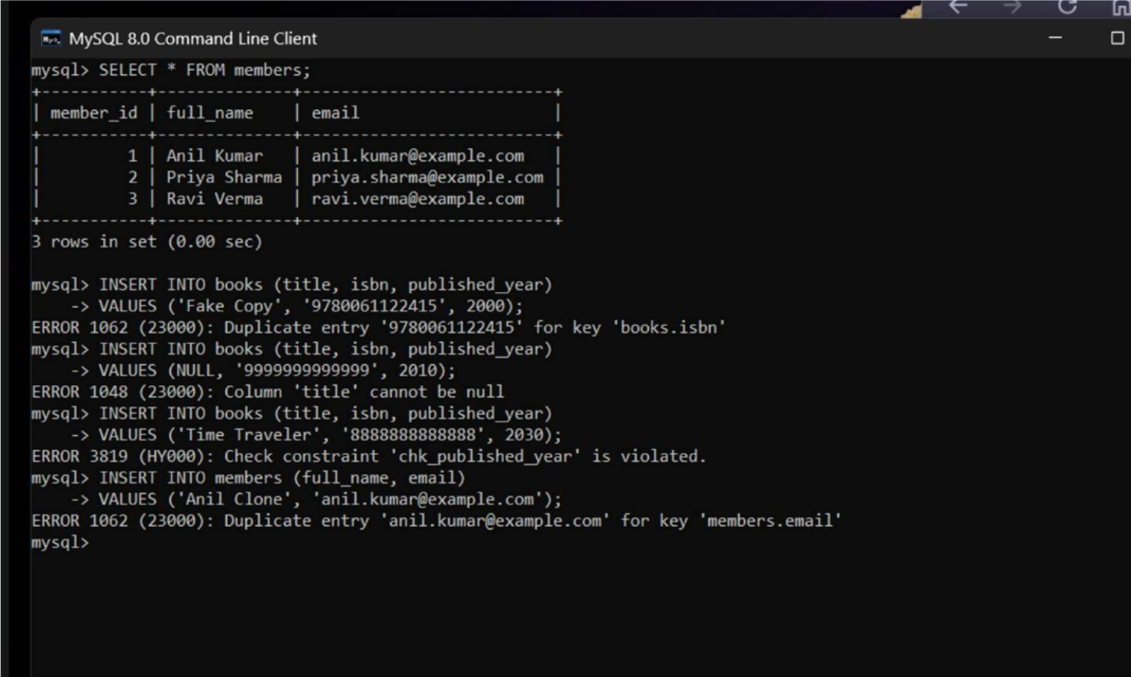
```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Explanation: 2030 fails the condition `published_year < 2027`, so the named CHECK constraint `chk_published_year` blocks the insert. Garbage dates cannot enter the table.

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Explanation: `anil.kumar@example.com` is already registered to `member_id 1`, so the UNIQUE constraint on email rejects the second signup. One email = one account



```
mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES (NULL, '99999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Time Traveler', '88888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql>
```